

FARHAN MOHAMMED

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Summary

I am an ML Engineer with over four years of experience in AI, ML, and software development. I have worked as an MLE specializing in NLP and large Language Models (LLMs), RAGs, and agents. I am available for immediate full-time work on CPT and will require H1-B sponsorship in the future.

Work Experience

Augmedix - Senior Natural Language Processing Engineer

San Francisco, California (Remote) | 02/2022 - 10/2024 – Ambient medical note creation using NLP in healthcare.

- Designed and developed a medical scribe AI agent to create a medical note with a 92% f1 score and <10% hallucination.
- Architected a hierarchical multi-agent system to extract different medical note sections, completing an entire note in less than 40s.
- AI agent achieved a 20% increase in accuracy and a 40% reduction in note completion time compared to the existing solution.
- Prompt engineering experience using techniques like COT, few shot, and prompt chaining.
- Deployed containerized cloud services to handle various aspects of the system, like transcription, model inference, and data storage.
- Designed cloud-based services to handle an ensemble of LLMs (GPT, med-PaLM, Gemini, Llama) having a <1% response failure rate.
- Optimized services by adding retry mechanisms, parallel and asynchronous model inference, and timeouts, decreasing runtime by 30%.
- Finetuned BERT models for relation extraction (82% acc), complaint classification (83% F1 score), and clinical NER (87% F1 score)
- Built multiple ML pipelines to pre-process the data, finetune, evaluate, and deploy models, handling over 10k medical notes biweekly
- Used Optimization techniques like 16-bit quantization for model inference (1.5x increase in speed) and LoRA for fine-tuning using PEFT
- Created a RAG-based system for extracting the top 5 ICD-10 medical codes for complaints from a directory of 70K codes.
- Achieved a hit rate of 95% for the RAG system and an average MRR of 0.79 (indicating the correct code present in the top 2 rankings)
- Assisted in hiring 6 MLEs and interviewed over 20 candidates for junior, senior, and principal roles

EPICS at ASU - Software Engineer

Phoenix, Arizona | 06/2021 - 02/2022 – Helping early-stage startup founders build their initial product

- Built UI components with Python and PyQt, integrating OpenCV for video and image processing.
- Developed an NLP-based algorithm matching users with companies based on their brand attributes.
- Achieved an 87% accuracy, matching over 500 companies across 10 different industries.
- Finetuned NER models for keyword extraction and matching and used the colBERT model for semantic text matching.

ITVersity, Inc. - Software Engineer Intern

Hyderabad, India | 01/2019 - 06/2019 – IT consulting firm

- Developed sentiment analysis model for a food delivery app with 91% accuracy using Python, Keras, and NLTK.
- Designed and implemented 10 UI components for a web app using React.js.
- Presented technical reports on model performance and API endpoints to stakeholders.

Projects

Mapping Healthcare Terms in Clinical Text to Standardized Medical Dictionary (UMLS)

Implemented a named entity linking model to map medical entities to over 400K unique UMLS identifiers. Used clinical BERT model to obtain entities. Extended the problem to create a ranking model using Rank Net to get an entity's top 3 UMLS codes.

Education

Trine University

Master of Business Administration - MBA, Project Management (Part-time) | 06/2024 - 05/2026

Arizona State University

Master of Science - MS, Computer Science | 01/2019 - 05/2021

Osmania University

Bachelor of Engineering - BE, Information Technology | 01/2015 - 12/2019

Skills

Natural Language Processing (NLP) - Machine Learning - Model building, training, and deployment - Prompt Engineering - Fine-tuning Generative AI
Deep learning frameworks - Data Analysis and Visualization - Full stack development

Large Language Models (LLMs) - Retrieval-Augmented Generation (RAG) - LangChain - Hugging Face Transformers - PyTorch - SpaCy - NLTK - NumPy - Pandas - scikit-learn - Vertex AI - OpenAI SDK - BERT - GPT - Gemini - Text Embeddings - Vector Databases

Programming Languages and Tools: Python - Java - JavaScript - AWS - GCP - SageMaker - Flask - SQL - Git - React.js - Agile